NETWORKWALKS — WEEK 02 CYBERSECURITY PRACTICAL TRAINING

PENETRATION TESTING REPORT

Footprinting, Reconnaissance & Network Scanning

Practical Security Assessment Report | Reference: W2-PM | Organization: NetworkWalks

Program	NetworkWalks — Week 02	**Assessment Reference**	W2-PM
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Organization	NetworkWalks	**Report Type**	Practical Security Assessment Report

1. DOCUMENT CONTROL

Attribute	Specification
Document Title	Penetration Testing Report — Footprinting, Reconnaissance & Network Scanning
Assessment Reference	W2-PM (NetworkWalks — Week 02)
Author / Assessor	Hafsa Khan
Reviewing Trainer	Waqas Karim (NetworkWalks)
Evidence Repository Reference	`github.com/Hafsa-Khan01/CyberForge-Labs/tree/main/Networkwalks/Week-02/Evidence`
Handling & Privacy Control	Private local IPv4 host identifiers and hardware MAC addresses are omitted/redacted.

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5	Scope of Assessment	**PM1, PM4, PM5**
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3. EXECUTIVE SUMMARY

This report presents the technical methodology, verified outputs, and security observations from the NetworkWalks Week 02 practical security assessment (Reference: **W2-PM**), prepared by **Hafsa Khan** under the supervision of trainer **Waqas Karim**. The engagement evaluated passive and active footprinting techniques, open-source intelligence (OSINT) collection, and local network host discovery across three assigned practical modules.

In **PM1 (Footprinting)**, external reconnaissance of **networkwalks.com** identified GoDaddy as the domain registrar, resolved the primary IPv4 A record to **192.232.216.135**, enumerated approximately eight DNS resource records across SOA, NS, MX, A, TXT, and SRV categories, and fingerprinted an Apache web server running **WordPress 7.1.4** with the **WP Download Manager** plugin behind a **ModSecurity (SpiderLabs)** Web Application Firewall. In **PM4 (theHarvester)**, automated OSINT enumeration against **microsoft.com** encountered upstream source-side errors and disconnections and is formally recorded as *Attempted — Output Not Validated*. In **PM5 (Zenmap)**, a non-intrusive ping scan across a local IPv4 /24 subnet (256 addresses) completed in approximately **13.84 seconds** and identified **2 active hosts**: a local **Huawei** router/gateway and a local **Windows** host.

Module	Target Scope	Assessment Technique	Validation Status
PM1	networkwalks.com	WHOIS, DNS Enumeration, HTTP Header & CMS/WAF Fingerprinting	**Completed — Verified**
PM4	microsoft.com	Passive OSINT Harvesting (theHarvester)	**Attempted — Output Not Validated**
PM5	Local IPv4 /24 Network	ICMP/ARP Ping Scan & Topology Mapping (Zenmap)	**Completed — Verified**

4. ASSESSMENT OBJECTIVES

The objective of Assessment W2-PM was to execute pre-exploitation reconnaissance and host discovery workflows in accordance with structured penetration testing standards:

- **External Infrastructure Profiling (PM1):** Gather domain registration metadata, authoritative DNS records, HTTP response headers, web application frameworks, and Web Application Firewall indicators for **networkwalks.com** without intrusive probing.
- **OSINT Collection Evaluation (PM4):** Assess automated public-source harvesting capabilities against **microsoft.com** using **theHarvester** and accurately document source-side operational constraints.
- **Local Network Host Discovery (PM5):** Perform an ICMP/ARP ping sweep across a local IPv4 /24 subnet using **Zenmap** to enumerate live hosts, identify gateway vendor hardware, and map logical network topology while maintaining privacy controls.
- **Defensive Posture Analysis:** Translate technical reconnaissance outputs into objective informational observations and practical hardening recommendations without overstating unverified risks.

5. SCOPE OF ASSESSMENT

Assessment activities were confined strictly to the targets and non-intrusive techniques specified in the NetworkWalks Week 02 practical curriculum:

Module	Target Asset	In-Scope Activities	Out-of-Scope Activities
PM1	networkwalks.com	WHOIS lookup, Nslookup, DNSRecon, cURL header inspection, WhatWeb CMS detection, Wafw00f WAF probing.	Vulnerability exploitation, brute-force attacks, directory fuzzing, denial-of-service testing.
PM4	microsoft.com	Passive public-source queries via theHarvester CLI.	Active scanning, credential testing, or probing of Microsoft infrastructure.
PM5	Local IPv4 /24 Subnet (256 Addresses)	Zenmap Ping Scan (host discovery) and visual topology mapping.	TCP/UDP port scanning, service version enumeration, OS fingerprinting, vulnerability scanning.

6. RULES OF ENGAGEMENT AND LIMITATIONS

This assessment was conducted as a structured practical security exercise under NetworkWalks Week 02 training guidelines. No commercial production penetration testing authorization beyond educational reconnaissance exercises is claimed. All testing adhered to the following operational boundaries and technical limitations:

- **Non-Intrusive Execution Only:** All external queries (PM1 and PM4) relied on standard public DNS/WHOIS protocols, passive OSINT aggregators, and standard HTTP/HTTPS GET/HEAD requests. Local network activity (PM5) was restricted exclusively to a ping scan.
- **No Vulnerability Exploitation or Port Scanning:** Because no port scans or vulnerability assessments were executed in PM1, PM4, or PM5, observations represent informational exposure and architectural indicators rather than confirmed exploitable vulnerabilities.
- **Third-Party OSINT Dependency (PM4):** Automated OSINT collection via **theHarvester** depends on external search engines and public APIs; upstream rate limiting and connection resets prevented result validation during PM4 execution.
- **DNSSEC Confirmation Boundary (PM1):** Automated DNS enumeration did not yield conclusive cryptographic validation of DNSSEC deployment; consequently, DNSSEC status is documented as unconfirmed rather than disabled.
- **Privacy & Data Sanitization (PM5):** Specific private RFC 1918 IPv4 addresses and hardware MAC addresses observed on the local laboratory network are intentionally omitted to protect environment confidentiality.

7. METHODOLOGY

The assessment followed a three-stage reconnaissance methodology aligned with industry-standard penetration testing frameworks (PTES and NIST SP 800-115 pre-attack phases):

1. **Phase 1 — Passive Footprinting & DNS Reconnaissance (PM1 & PM4):** Collected domain registration ownership data via WHOIS, queried authoritative DNS zones across standard resource record types, and attempted passive OSINT aggregation from external search providers.
2. **Phase 2 — Non-Intrusive Web & Perimeter Fingerprinting (PM1):** Inspected HTTP/HTTPS server response headers, identified Content Management System (CMS) and plugin signatures, and evaluated perimeter Web Application Firewall responses.
3. **Phase 3 — Active Local Host Discovery & Topology Mapping (PM5):** Executed an ICMP/ARP ping sweep across the local IPv4 /24 subnet to identify responsive nodes, resolve hardware vendor OUIs, and construct a logical topology view.

8. TOOLS AND TECHNOLOGIES

Tool	Category	Module	Technical Function in Assessment
WHOIS	Registry Lookup	PM1	Queried domain registrar attribution, registration dates, and technical contact metadata.
Nslookup	DNS Query Utility	PM1	Performed targeted DNS resolution for A, AAAA, MX, NS, and TXT resource records.
cURL	HTTP Client	PM1	Issued HTTPS requests to verify HTTP status codes and inspect raw server response headers.

Tool	Category	Module	Technical Function in Assessment
WhatWeb	Web Fingerprinter	PM1	Identified CMS platform versions, active plugins, and supporting web technologies.
Wafw00f	WAF Detection	PM1	Analyzed HTTP response signatures to fingerprint Web Application Firewall protections.
DNSRecon	DNS Enumeration	PM1	Conducted multi-record DNS zone enumeration across SOA, NS, MX, A, TXT, and SRV records.
theHarvester	OSINT Harvester	PM4	Attempted passive collection of public emails, subdomains, and host records from external sources.
Zenmap	Network Mapper	PM5	Executed a local IPv4 /24 ping scan to detect live hosts, gateway vendor, and network topology.

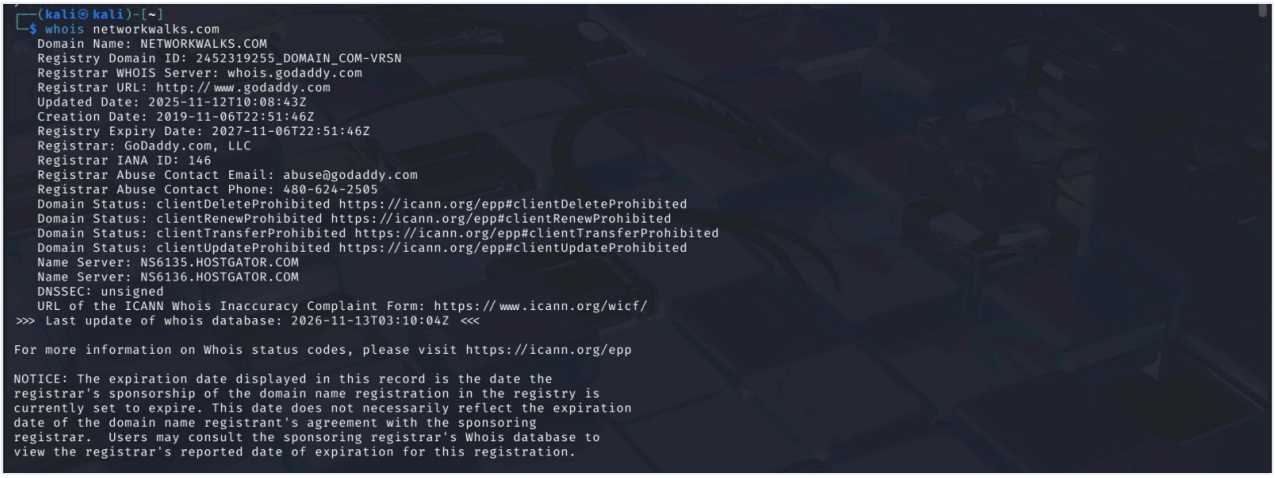
9. TECHNICAL ASSESSMENT RESULTS

9.1 PM1 — Footprinting (Target: networkwalks.com)

Footprinting of **networkwalks.com** combined registry, DNS, and HTTP-layer inspection tools to establish a verified baseline of external infrastructure exposure.

- **Domain Registration (WHOIS):** WHOIS queries confirmed that **networkwalks.com** is registered through **GoDaddy**. Registration lifecycle timestamps and technical registrar metadata were reviewed.
- **Targeted DNS Resolution (Nslookup):** Nslookup queries resolved the primary IPv4 **A record to 192.232.216.135**. Additional query activity was executed across **AAAA, MX, NS, and TXT** record categories to inspect IPv6, mail routing, name server delegation, and domain verification records.
- **Comprehensive DNS Enumeration (DNSRecon):** DNSRecon enumerated approximately **eight (8) DNS records** spanning **SOA, NS, MX, A, TXT, and SRV** categories. DNSSEC deployment status could not be confirmed from the enumeration output and is therefore recorded as unconfirmed.
- **HTTP Header Inspection (cURL):** Direct HTTPS requests returned an **HTTP 200 OK** response status accompanied by **Apache-related server headers**.
- **Web Technology Fingerprinting (WhatWeb):** WhatWeb identified the underlying Content Management System as **WordPress version 7.1.4** alongside supporting technology indicators, including the **WP Download Manager** plugin.
- **Web Application Firewall Detection (Wafw00f):** Wafw00f analysis identified a **ModSecurity (SpiderLabs)** WAF fingerprint protecting the web application tier.

Reconnaissance Parameter	Verified Output / Finding	Verification Tool
Domain Registrar	GoDaddy (registration & technical metadata reviewed)	WHOIS
Primary IPv4 Address (A Record)	192.232.216.135	Nslookup / DNSRecon
Queried DNS Record Types	SOA, NS, MX, A, AAAA, TXT, SRV (~8 total records enumerated)	Nslookup / DNSRecon
DNSSEC Deployment Status	Not confirmed (neither claimed enabled nor disabled)	DNSRecon
HTTP Status & Web Server	HTTPS HTTP 200 OK; Apache-related response headers	cURL
CMS & Plugin Stack	WordPress version 7.1.4; WP Download Manager	WhatWeb
Web Application Firewall	ModSecurity (SpiderLabs) fingerprint detected	Wafw00f



Technical Explanation: Displays the initial WHOIS query execution against `networkwalks.com` confirming domain registration status and registrar identification as GoDaddy.

Figure 1: WHOIS domain registration lookup for networkwalks.com identifying GoDaddy as the registrar.



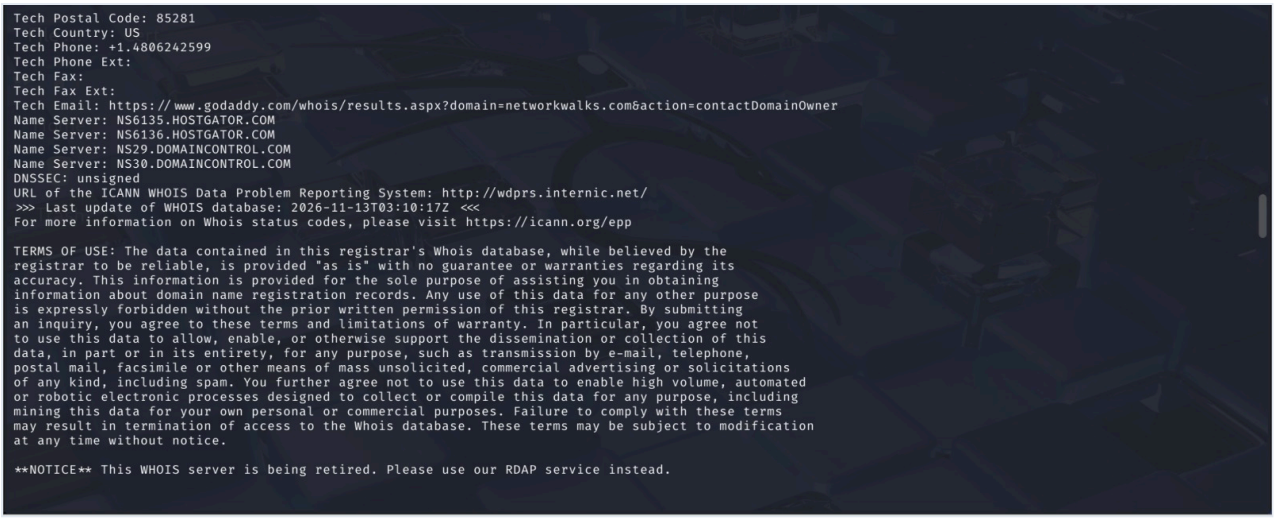
Technical Explanation: Captures registrar registration details including domain status flags, creation/expiration metadata, and registrar abuse contact fields.

Figure 2: WHOIS registration details showing domain lifecycle metadata and registrar attribution.



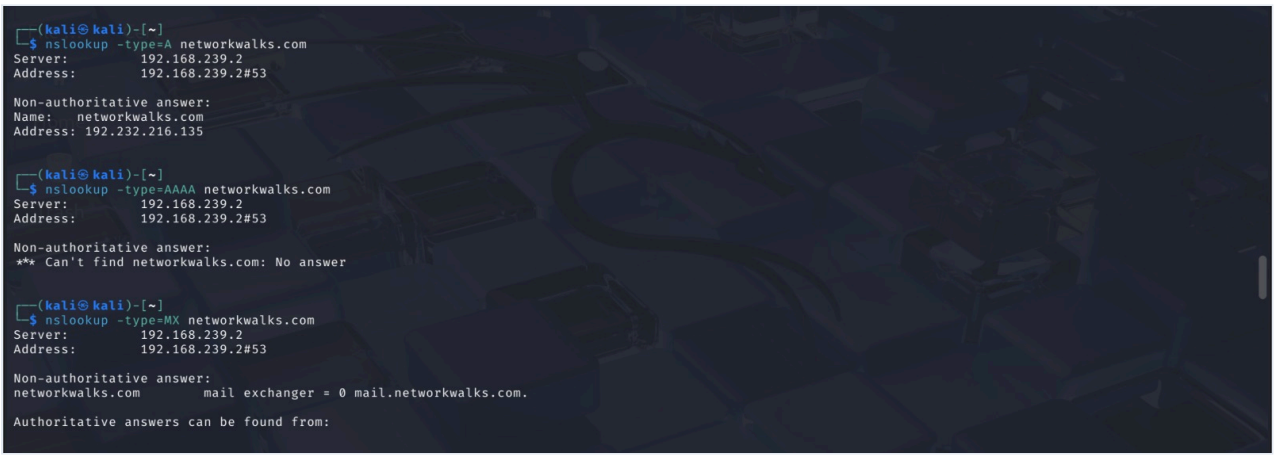
Technical Explanation: Documents technical WHOIS contact and name server delegation metadata associated with the GoDaddy registration record.

Figure 3: WHOIS technical metadata detailing authoritative name server references and registry fields.



Technical Explanation: Presents the concluding segment of WHOIS domain records and registry notice statements reviewed during passive footprinting.

Figure 4: WHOIS domain records summary completing the passive registry review for networkwalks.com.



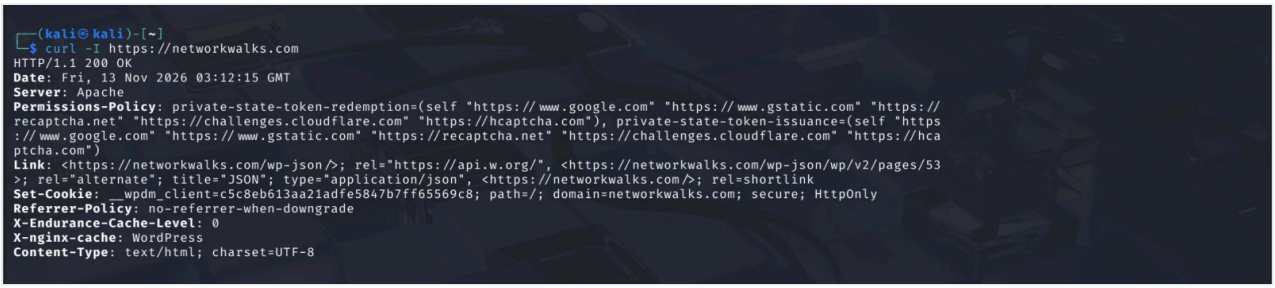
Technical Explanation: Shows Nslookup resolving the primary IPv4 A record for `networkwalks.com` to `192.232.216.135`.

Figure 5: Nslookup DNS query output confirming IPv4 A record resolution to 192.232.216.135.



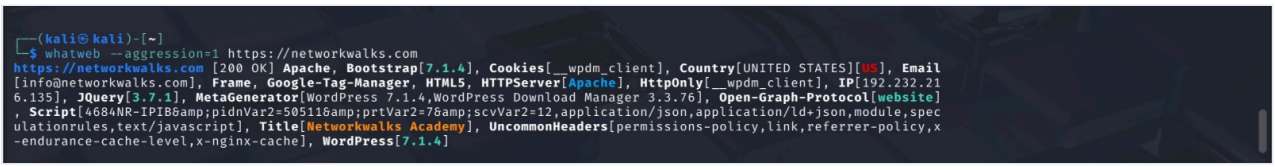
Technical Explanation: Illustrates targeted Nslookup queries performed across AAAA, MX, NS, and TXT record types to verify mail exchange and name server configurations.

Figure 6: Nslookup record query activity across AAAA, MX, NS, and TXT DNS categories.



Technical Explanation: Captures cURL HTTPS header retrieval returning HTTP 200 OK along with Apache-related web server response headers.

Figure 7: cURL HTTPS response header inspection showing HTTP 200 OK and Apache-related headers.



Technical Explanation: Displays WhatWeb technology fingerprinting results identifying WordPress version 7.1.4 on `networkwalks.com`.

Figure 8: WhatWeb technology detection output reporting WordPress version 7.1.4.



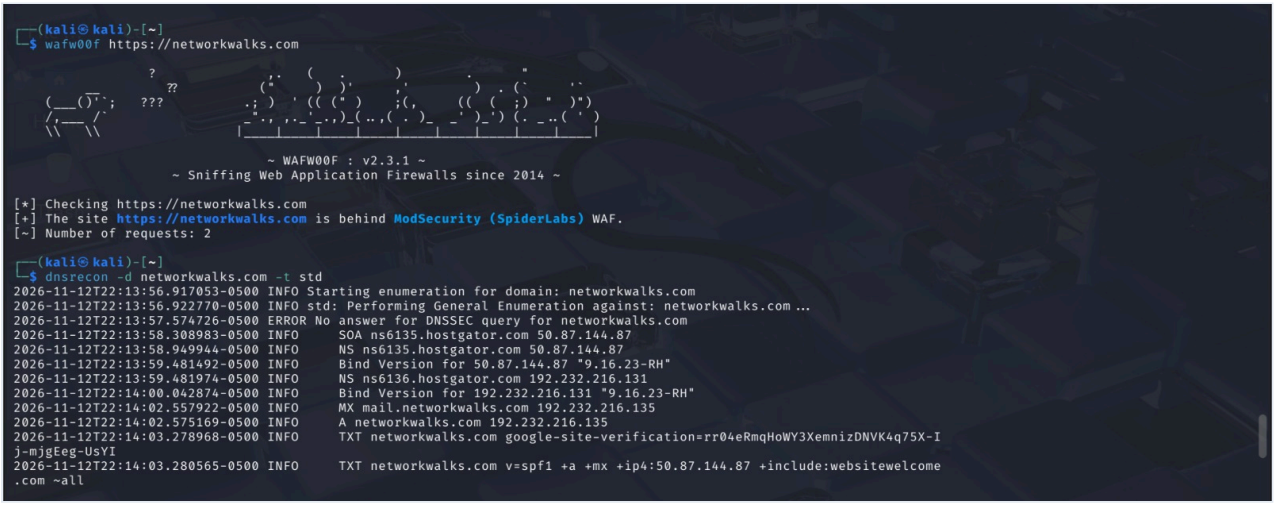
Technical Explanation: Documents extended WhatWeb plugin and HTTP metadata output, including detection of the WP Download Manager plugin.

Figure 9: WhatWeb additional fingerprinting output identifying WP Download Manager and supporting indicators.



Technical Explanation: Shows Wafw00f HTTP probe execution detecting a ModSecurity (SpiderLabs) Web Application Firewall signature.

Figure 10: Wafw00f firewall detection output identifying a ModSecurity (SpiderLabs) fingerprint.



Technical Explanation: Summarizes DNSRecon zone enumeration identifying approximately eight DNS records across SOA, NS, MX, A, TXT, and SRV categories, with DNSSEC status remaining unconfirmed.

Figure 11: DNSRecon enumeration results detailing ~8 records across SOA, NS, MX, A, TXT, and SRV types.

9.2 PM4 — theHarvester (Assigned Target: microsoft.com)

In Module PM4, **theHarvester** was executed against the assigned target **microsoft.com** to evaluate passive OSINT enumeration of public email addresses, subdomains, and host records from external search engines and data aggregators.

- **Execution Status: Attempted — Output Not Validated**.
- **Operational Limitation:** During execution, external data providers returned source-side errors and connection resets/disconnections that interrupted automated harvesting before reliable results could be retrieved or verified.
- **Reporting Integrity:** In strict compliance with assessment verification standards, no unverified email addresses, subdomains, hostnames, IP addresses, or discovery counts are claimed or extrapolated from this incomplete run.



Technical Explanation: Captures the initial command invocation of `theHarvester` targeting `microsoft.com` across public OSINT data sources.

Figure 12: theHarvester enumeration attempt initiated against microsoft.com (Status: Attempted — Output Not Validated).



Technical Explanation: Documents the upstream source-side errors and network disconnections encountered during query execution that prevented output validation.

Figure 13: theHarvester execution log showing source-side errors and disconnections during OSINT queries.



Technical Explanation: Shows the CLI parameter syntax and source configuration reference verified during troubleshooting of theHarvester execution.

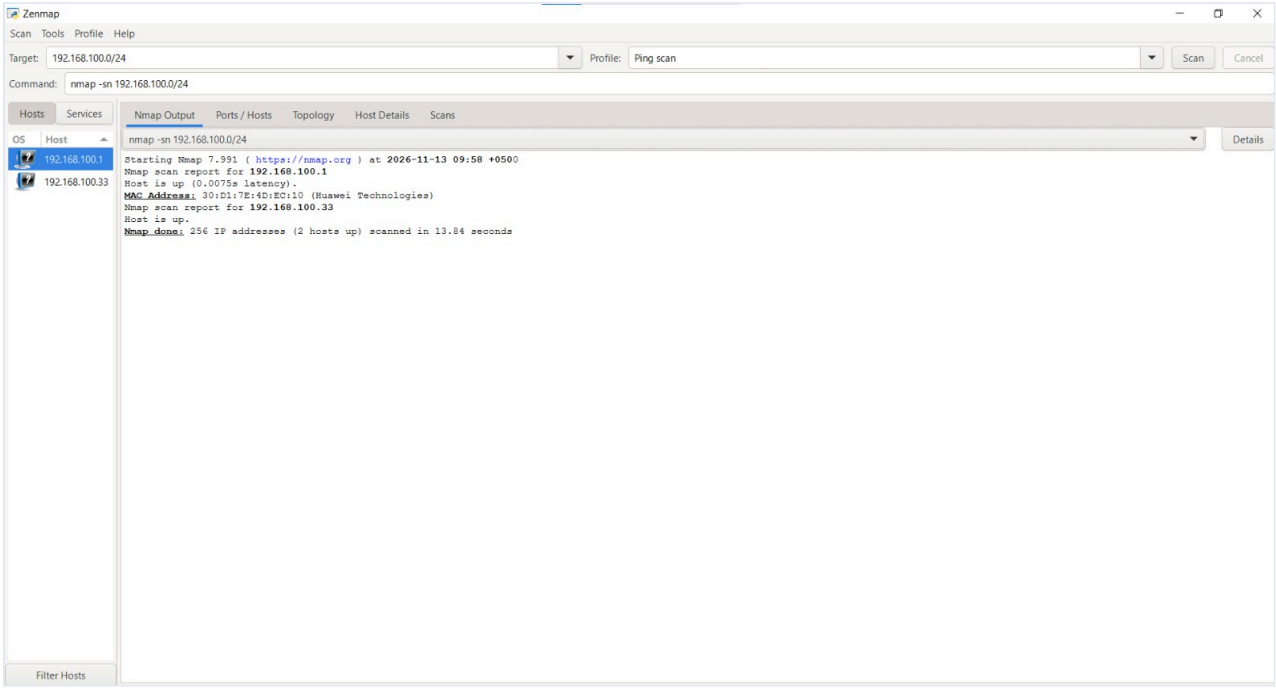
Figure 14: theHarvester command reference and module options reviewed during troubleshooting.

9.3 PM5 — Zenmap (Activity: Local Network Host Discovery)

In Module PM5, **Zenmap** was used to perform non-intrusive host discovery across the local IPv4 /24 network segment. Specific private IPv4 addresses and MAC hardware identifiers are withheld from this report to preserve local network privacy.

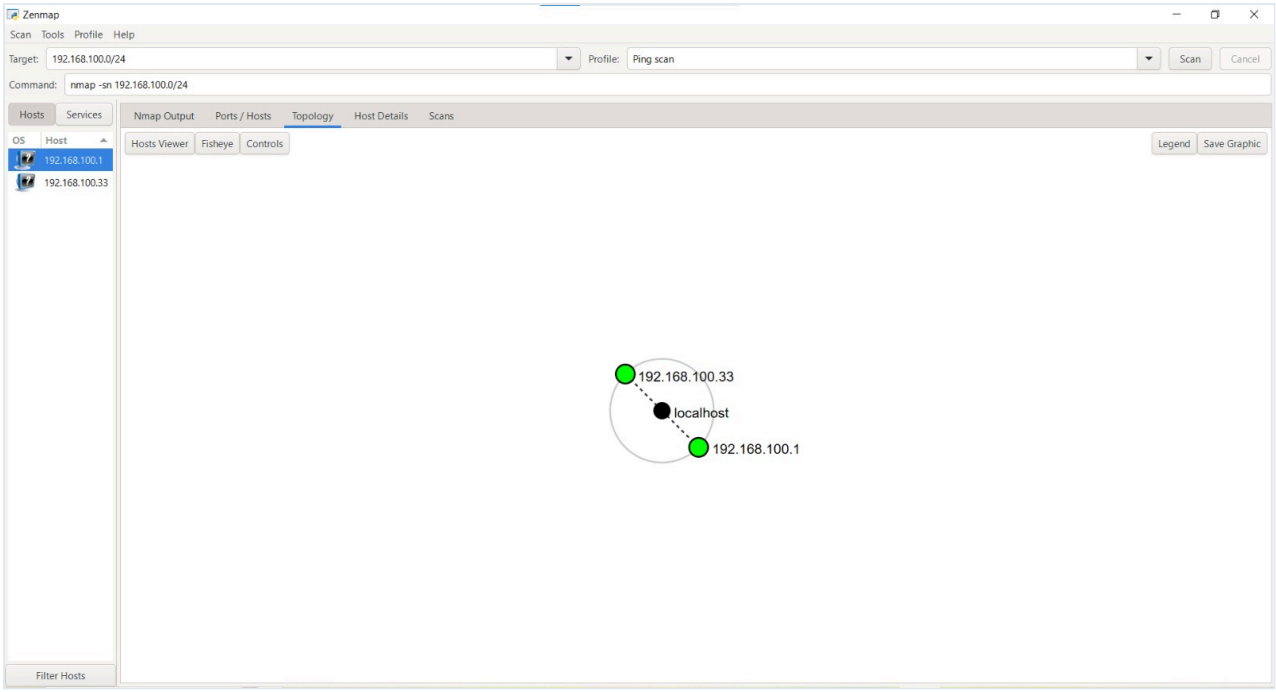
- **Scan Profile & Range:** Configured as a **Ping Scan** covering a **local IPv4 /24** subnet (**256 IP addresses scanned**).
- **Discovery Metrics:** Out of 256 scanned addresses, exactly **2 hosts were reported up**. The scan completed in approximately **13.84 seconds**.
- **Topology & Hardware Vendor Identification:** Zenmap's topology view mapped two connected nodes: the local router/gateway (reported vendor: **Huawei**) and a local **Windows host**.
- **Scope Boundary Clarification:** Because only an ICMP/ARP ping scan was executed, no TCP/UDP port scanning, service enumeration, exploitation, or vulnerability discovery was conducted.

Scan Parameter	Observed Output
Scanning Tool & Profile	Zenmap — Ping Scan (Host Discovery Only)
Target Network Range	Local IPv4 /24 Subnet (256 IP addresses scanned)
Responsive Hosts	2 hosts reported up (private IPs and MACs redacted)
Scan Duration	Approximately 13.84 seconds
Topology & Vendor Identification	Local Router/Gateway (Vendor: Huawei) and Windows Host
Port / Vulnerability Enumeration	None (not within scope of a ping scan)



Technical Explanation: Displays Zenmap ping scan output confirming 256 IP addresses scanned across the local IPv4 /24 range, 2 hosts reported up, Huawei gateway vendor identification, and an elapsed time of approximately 13.84 seconds.

Figure 15: Zenmap ping scan results showing 256 addresses scanned, 2 responsive hosts, Huawei router vendor, and ~13.84s duration.



Technical Explanation: Illustrates Zenmap's graphical network topology view mapping the connection between the Huawei local router/gateway and the Windows host.

Figure 16: Zenmap network topology view depicting the Huawei local router/gateway and Windows host.

10. FINDINGS AND OBSERVATIONS

The observations below summarize the technical findings from PM1, PM4, and PM5. Consistent with evidence-based reporting standards, all items are classified strictly as **Informational Observations** or **Operational Limitations**; no unverified CVSS scores, severity ratings, or exploitable vulnerability claims are assigned without proof.

ID	Observation Title	Classification	Technical Details & Analytical Context
OBS-01	Public DNS and Registration Information Exposure (PM1)	Informational Observation	WHOIS, Nslookup, and DNSRecon publicly exposed GoDaddy registrar metadata, IPv4 hosting resolution (`192.232.216.135`), and ~8 DNS records across SOA, NS, MX, A, TXT, and SRV types. DNSSEC status could not be confirmed from enumeration outputs.
OBS-02	Web Technology Fingerprinting Exposure (PM1)	Informational Observation	cURL and WhatWeb disclosed Apache-related HTTP response headers, **WordPress version 7.1.4**, and the **WP Download Manager** plugin. While normal for public web hosting, explicit software version banners reduce reconnaissance effort for external actors.
OBS-03	WAF Fingerprint Observation (PM1)	Informational Observation (Defensive Control)	Wafw00f detected a **ModSecurity (SpiderLabs)** signature on `networkwalks.com`, confirming the presence of an active application-layer filtering control in front of the WordPress instance.

ID	Observation Title	Classification	Technical Details & Analytical Context
OBS-04	theHarvester Execution Limitation (PM4)	Operational Limitation (Not Validated)	OSINT enumeration targeting `microsoft.com` encountered upstream source-side errors and disconnections, preventing reliable validation. This highlights the operational dependency of automated OSINT tools on third-party source availability and rate limits.
OBS-05	Two Responsive Hosts Identified on Local Network (PM5)	Informational Observation	A Zenmap ping scan across the local IPv4 /24 network (256 addresses) discovered 2 active hosts (a Huawei router/gateway and a Windows host) in ~13.84 seconds, demonstrating standard ICMP/ARP responsiveness on the local segment.

11. RISK AND SECURITY RECOMMENDATIONS

The following practical, technically grounded recommendations correspond directly to each documented observation:

Ref	Related Finding	Recommended Technical Action
REC-01	OBS-01 (DNS & Registration Exposure)	Maintain registrar privacy protection on GoDaddy WHOIS contact records, audit the ~8 public DNS records (SOA, NS, MX, A, TXT, SRV) to ensure no stale entries remain, and verify DNSSEC zone signing configuration with the DNS provider.
REC-02	OBS-02 (Web Technology Fingerprinting)	Harden Apache HTTP response headers (e.g., setting `ServerTokens Prod` and `ServerSignature Off`) and suppress explicit WordPress core and WP Download Manager version metadata in public page headers and generator tags.
REC-03	OBS-03 (ModSecurity WAF Presence)	Retain ModSecurity (SpiderLabs) in active blocking mode, keep OWASP Core Rule Set (CRS) definitions current, and review WAF logs periodically to monitor automated reconnaissance and probe traffic.
REC-04	OBS-04 (OSINT Execution Limitation)	Supplement automated OSINT tools like `theHarvester` with API-authenticated data sources and manual multi-source verification when public search engines enforce rate limits or drop connections, and always mark incomplete runs as unvalidated.
REC-05	OBS-05 (Local Network Host Discovery)	Ensure local network segmentation isolates administrative gateway interfaces on the Huawei router, restrict ICMP echo responses where appropriate, and maintain active host-based firewall rules on the Windows endpoint.

12. CONCLUSION

Assessment **W2-PM** successfully demonstrated structured footprinting, OSINT collection, and local network host discovery within authorized non-intrusive boundaries. External reconnaissance of **networkwalks.com** (PM1) accurately established domain registration via GoDaddy, IPv4 hosting at **192.232.216.135**, approximately eight DNS records, Apache/WordPress 7.1.4/WP Download Manager technology signatures, and ModSecurity WAF protection. Passive OSINT harvesting against **microsoft.com** (PM4) encountered upstream source disconnections and was transparently documented as *Attempted — Output Not Validated*. Local network host discovery (PM5) mapped two active nodes—a Huawei router/gateway and a Windows host—across a 256-address IPv4 /24 range in approximately 13.84 seconds while safeguarding private network identifiers.

13. EVIDENCE APPENDIX

Complete manifest of the 16 practical evidence screenshots referenced in this assessment from the CyberForge-Labs repository (`Networkwalks/Week-02/Evidence`):

Figure	Module Folder	Evidence Filename	Verification Subject
Figure 1	PM1-Footprinting	`01-whois-domain-registration.jpeg`	WHOIS domain registration & GoDaddy registrar
Figure 2	PM1-Footprinting	`02-whois-registration-details.jpeg`	WHOIS registration lifecycle metadata
Figure 3	PM1-Footprinting	`03-whois-technical-details.jpeg`	WHOIS technical & name server metadata
Figure 4	PM1-Footprinting	`04-whois-domain-records.jpeg`	WHOIS domain records summary
Figure 5	PM1-Footprinting	`05-nslookup-dns-query.jpeg`	Nslookup A record resolution (192.232.216.135)
Figure 6	PM1-Footprinting	`06-nslookup-dns-records.jpeg`	Nslookup AAAA, MX, NS, and TXT queries
Figure 7	PM1-Footprinting	`07-curl-http-response-headers.jpeg`	cURL HTTPS 200 OK & Apache headers
Figure 8	PM1-Footprinting	`08-whatweb-technology-detection.jpeg`	WhatWeb WordPress 7.1.4 detection
Figure 9	PM1-Footprinting	`09-whatweb-additional-output.jpeg`	WhatWeb WP Download Manager detection
Figure 10	PM1-Footprinting	`10-wafw00f-firewall-detection.jpeg`	Wafw00f ModSecurity (SpiderLabs) fingerprint

Figure	Module Folder	Evidence Filename	Verification Subject
Figure 11	PM1-Footprinting	`11-dnsrecon-dns-enumeration.jpeg`	DNSRecon ~8 records (SOA, NS, MX, A, TXT, SRV)
Figure 12	PM4-theHarvester	`01-theharvester-enumeration-attempt.jpeg`	theHarvester enumeration attempt on microsoft.com
Figure 13	PM4-theHarvester	`02-theharvester-source-errors.jpeg`	theHarvester source-side errors & disconnection
Figure 14	PM4-theHarvester	`03-theharvester-command-reference.jpeg`	theHarvester CLI command reference
Figure 15	PM5-Zenmap	`01-zenmap-ping-scan-results.jpeg`	Zenmap ping scan (256 IPs, 2 up, 13.84s, Huawei)
Figure 16	PM5-Zenmap	`02-zenmap-network-topology.jpeg`	Zenmap topology (Huawei gateway & Windows host)

14. FINAL ASSESSMENT NOTES

- **Evidence Integrity:** All technical claims in this report reflect strictly verified practical outputs. Unvalidated outputs (PM4) and unconfirmed parameters (DNSSEC status in PM1) are explicitly labeled without speculation.
- **Scope Discipline:** No port scanning, service exploitation, or vulnerability claims are made for PM5 or external targets, maintaining alignment with the non-intrusive ping scan and passive reconnaissance scope.
- **Portfolio & Submission Readiness:** Structured for formal submission to NetworkWalks (Trainer: Waqas Karim) and publication in the `Hafsa-Khan01/CyberForge-Labs` GitHub repository.